

Formulating Absolute Value Linear Equations

TEACHER KEY | Algebra II Honors | Unit 2 | Day 4 | TEK A2.6.D

Objective: I can formulate an absolute value linear equation from a real-world description, including situations that need an extra structural step beyond the basic template.

Vocabulary

Formulate — set up an equation from a description without solving it.

Center value / margin — the basic template is $|x - \text{center}| = \text{margin}$ - but not every situation hands you those two numbers directly.

Part 1 — Pull Out Center and Margin Yourself

Clarissa wants a house within 5 houses of house #1248 on an even-numbered street.

Center: **1248** Margin: **5** Equation: **$|x - 1248| = 5$**

A surveying instrument measures a 10-mile stretch with a 1.5% margin of error.

The 1.5% is a percent of the true distance, not a flat number of miles - it has to be built into the equation as a percent expression.

Equation: **$1.5 = |x - 10|/10 \cdot 100$**

Part 2 — A Layered Scenario: Brighton Builders' Hallway

A hallway has offices on both sides, evenly spaced from the center at 15x feet each way, and the total layout must equal 292 feet with 52 feet reserved for walls and doors.

This does not fit $|x - \text{center}| = \text{margin}$ directly - there is no subtraction from a center value. Identify what the absolute value expression represents here, and why.

Sample: **$|15x|$ represents the offices' total reach from the center in either direction - absolute value is used because the layout is symmetric, not because something is being compared to a target**

Equation: **$2|15x| + 52 = 292$**

Part 3 — Going Further: Generalize the Template

In your own words, explain what "center" and "margin" mean in a real-world $|x - \text{center}| = \text{margin}$ equation, and give one example of a situation where the equation does NOT fit that template (like Part 2).

Sample: **center is the target value everything is measured against; margin is the allowed distance from it; Part 2 does not fit because there is no target being compared against - the absolute value there measures symmetric spacing instead**

Part 4 — A Second Layered Scenario

A garden bed is built symmetric about a center walkway, with plant rows evenly spaced 4x feet from the walkway on each side, and the total bed width (rows plus a 3-foot walkway) is 27 feet.

Formulate the equation, and explain why it needs the layered approach rather than the basic template.

Equation: **$2|4x| + 3 = 27$ symmetric spacing from a walkway, not a target-and-tolerance comparison**

Part 5 — Classify Before You Formulate

For each scenario, decide: does it fit the basic $|x - \text{center}| = \text{margin}$ template, or does it need the layered approach?

Scenario	Template or Layered?
A part must weigh 12 oz, +/- 0.5 oz	Template - 12 is the target
A bridge has supports evenly spaced from its center, total span 200 ft	Layered - symmetric spacing, no target
A recipe needs 3 cups of sugar, within 1/4 cup	Template - 3 is the target
A picture frame border is the same width on all sides, total outer width 24 in.	Layered - symmetric border, no target

Practice — Formulate, Do Not Solve

1. A shipping company's box-weight target is 50 lb, with a tolerance of 3 lb. $|x - 50| = 3$
2. A factory's bolt length must be within 0.02 cm of 6 cm. $|x - 6| = 0.02$
3. A rectangular sign is built symmetric about its center line, with total width 40 in. and each side extending 2x inches from center. Formulate, and explain why this is like Part 2, not Part 1.

Equation: $2|2x| = 40$ symmetric spacing, not a target-and-tolerance situation

4. A quality-control spec allows a part's diameter to differ from 3 cm by at most 0.1 cm. Formulate and classify. $|x - 3| = 0.1$ Template
5. A fence is built symmetric about a center gate, with 6x feet of fencing on each side, and the total length including a 4-foot gate is 40 feet. Formulate and classify.

Equation: $2|6x| + 4 = 40$ Layered - symmetric spacing, no target value

Exit Ticket

A thermostat targets 70°F with an acceptable range of 2°F either way. Formulate the equation, then state whether it fits the basic template or needs the Part 2 approach, and why.

Equation: $|x - 70| = 2$ fits the basic template - 70 is a target value and 2 is the allowed tolerance from it